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Association of ADHD/ASD traits with premenstrual dysphoric disorder among full-time employed women in Japan: A cross-sectional study

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ABSTRACT

Objectives: Premenstrual dysphoric disorder (PMDD) has a substantial impact on adult female workers. This study investigates the relationship between attention-deficit/hyperactivity disorder (ADHD) and autism spectrum disorder (ASD) traits and PMDD among female employees with no history of pregnancy and not using contraceptives.

Methods: This cross-sectional study analyzed data from an online survey conducted in October 2023 among Japanese full-time female employees aged 20–44 who have not used contraceptives and who have not experienced pregnancy. ADHD/ASD traits were assessed using the Adult ADHD Self-Report Scale and the 10-item Autism Spectrum Quotient, respectively. We used the PMDD scale, which was developed in Japanese based on the research criteria described in Diagnostic and Statistical Manual of Mental Disorder-IV-text revision. Multiple logistic regression was conducted by adjusting age, educational attainment, and occupation.

Results: A total of 2000 participants were included in the analysis, of which 8.6 % exhibited ADHD traits, and 12.8 % exhibited ASD traits. The prevalence of PMDD was 5.1 %. Participants with ADHD traits had a significantly higher risk of PMDD, which remained significant after adjusting for covariates (adjusted OR = 6.49, 95 % CI: 4.10–10.2, $p < 0.001$). No significant association was found between ASD traits and PMDD.

Conclusion: Among full-time female employees, ADHD traits were significantly associated with suffering from PMDD, while no significant association was observed between ASD traits and PMDD. These present epidemiological results may contribute to improving workplace and clinical interventions to address PMDD.

1. Introduction

Menstrual health has historically been framed primarily as a sanitation issue, with insufficient attention and prioritization given to its broader implications. This neglect poses a global challenge, particularly in relation to women's mental health (Khofi, 2024). Premenstrual dysphoric disorder (PMDD), one severe form of premenstrual disorders, represents a significant public health concern, critically affecting the quality of life and daily functioning of women. The prevalence of PMDD is substantial, with current estimates suggesting that 3–9 % of women of reproductive age meet the stringent diagnostic criteria for the disorder (Robinson and Ismail, 2015). Beyond the inherent burden of PMDD, the disorder contributes to a considerable loss in disability-adjusted life years (DALYs), comparable to major recognized disorders (Halbreich et al., 2003). It also has been associated with work productivity impairment and increased absenteeism, affecting the working population (Heinemann et al., 2010). Furthermore, the prevalence of PMDD in women who have attempted suicide has been reported to be as high as 23 % (Ducasse et al., 2016). Given its substantial impact on women of reproductive age, PMDD is recognized as a condition that adversely affects both daily activities and workplace productivity for a significant number of women (Chawla et al., 2002; Loukzadeh et al., 2024).

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While multiple factors have been implicated in the development of PMDD, developmental disorders, such as attention deficit hyperactivity disorder (ADHD) and autism spectrum disorder (ASD), are considered to be some of the potential associated factors. The reported ADHD and ASD prevalence is increasing globally; ADHD affects 5–8 % of youth (Salari et al., 2023) and 2.5–7 % of adults (Song et al., 2021; Abdelnour et al., 2022), while ASD has a global prevalence of 0.6 % (Zeidan et al., 2022), increasing to approximately 2.2 % among adults in the United States (Hirota and King, 2023).To date, two previous studies reported that women with ADHD showed a higher prevalence of PMDD compared to women without ADHD. One cross-sectional study from the Netherlands pointed out that among 209 outpatient women aged 18–71 years old, those who were diagnosed with ADHD had a higher prevalence risk of having PMDD (Dorani et al., 2021). Another prospective study in Taiwan revealed that among 108 women who have regular menstrual cycles, those with PMDD were more likely to have ADHD traits and to have higher levels of inattention during the menstrual cycle compared to the control group (Lin et al., 2024). Otherwise, one study did not find an increased prevalence of PMDD in ASD (Groenman et al., 2022), while another study reported that the prevalence of premenstrual syndrome (PMS) was significantly high in ASD (Obaydi and Puri, 2008). The results are still inconsistent, and empirical studies on this topic remain scarce.

Many of these studies are constrained by their focus on specific regions or smaller numbers of people with a wider range of age, limiting their generalizability to the broader female workforce. Moreover, although ADHD and ASD share environmental and biological risk factors (Antshel et al., 206), which may synergistically influence menstrual symptoms in women through similar mechanisms, few studies simultaneously examine how ADHD and ASD are associated with PMDD. In addition, the number of participants in previous studies was small, and the association between developmental traits and PMDD is still unclear. In the previous study many of the people were already using low-dose contraceptives, which contributes to the treatment of PMDD. In Japan, the adoption of contraceptive pills is low, which presents a different pattern from other countries (Yoshida et al., 2016). The previous studies also did not differentiate between the history of pregnancy. However the hormonal fluctuations in pregnancy can drive a neurophysiological change of mood, which should be distinguished from mental health issues associated with PMDD (Grattan and Ladyman, 2020). Thus, the association between developmental traits and PMDD should be assessed under conditions that are not affected by medication or pregnancy.

PMDD is a multifaceted condition influenced by various risk factors, including genetic predispositions (Yonkers et al., 2008), biological factors such as age at menarche, and psychological factors including self-reported depression (Trieu et al., 2023). Lifestyle factors, including dietary habits (MoradiFili et al., 2020), physical exercise (Liguori et al., 2023), caffeine consumption (Wu et al., 2024), and sleep disturbance (Jehan et al., 2016) also contribute to its development. Therefore, we examined the interaction between ADHD/ASD traits and lifestyle ormedication-related factors while accounting for basic lifestyle and medication-related factors. By examining the relationship between ADHD/ASD traits and PMDD from modifiable factors and by focusing specifically on the Asian workforce, this research seeks to identify variables that are specifically effective for developmental traits that may provide insights into strategies for both the prevention and management of PMDD.

This study aimed to investigate the association between ADHD/ASD traits and the presence of PMDD among female employees who have not experienced pregnancy and have not used contraceptives. We assessed whether PMDD status is associated with existence of developmental traits, hypothesizing that ADHD/ASD traits have positive association between PMDD. Additionally, we examined the interaction effect of ADHD/ASD traits and lifestyle or medication-related factors on PMDD.

2. Methods

2.1. Study design and setting

This cross-sectional study utilized data from an online survey conducted in August 2023 among full-time female employees in Japan (N = 2000). The survey contains information on multiple aspects including menstrual symptoms, developmental traits, lifestyle patterns, and medication-related behaviors. The research protocol for this study was approved by the ethics committee of the Graduate School of Medicine and Faculty of Medicine at The University of Tokyo (Approval No. 2023058NI-(1)).

2.2. Participants

Participants were selected from the registered panel members of an online survey company, Cross Marketing Inc. (n.d.). To be eligible for this study, participants had to meet the following criteria: (1) be Japanese women, (2) between 20 and 44 years old, (3) have no history of pregnancy, (4) be employed full-time, and (5) have not used contraceptives in the past year. No exclusion criteria were applied. Participants were recruited from the registered panel of an online survey company. The panel comprised over 13 million individuals with diverse sociodemographic backgrounds, ensuring national representation. Based on pre-registered demographic data, including gender and age, the survey company sent invitation emails to individuals who were most likely to meet the study's eligibility criteria. Those who agreed to participate after reviewing detailed study information and meeting the eligibility requirements completed the survey. Panelists were free to skip any part of the questionnaire or discontinue their participation at any time. The survey concluded once the target number of 2000 respondents was reached. To ensure data integrity, the online survey system was designed to prevent repeated or duplicated submissions from the same respondent. Participants were unable to proceed with the survey if duplication was detected by the system. In line with established quality control protocols, panel registration using identical personal information is strictly prohibited. Additionally, routine data cleaning procedures are implemented to identify and remove duplicate entries based on overlapping personal information or identical email addresses.

2.3. Measurement variables

2.3.1. Premenstrual dysphoric disorder (PMDD)

Participants were asked in the questionnaire about their menstrual symptoms. We used the PMDD scale, which was developed in Japanese based on the research criteria described in Diagnostic and Statistical Manual of Mental Disorder-IV-text revision (DSM-IV-TR) (Cooper, 2001). This confirmed its reliability and validity (Miyaoka et al., 2009; Lewis, 1996). The PMDD scale mainly consists of two parts. Section I includes 12 items, assessing the respondent's PMDD symptom severities on a scale of 1–4 (1 = *none*, 2 = *mild*, 3 = *moderate*, 4 = *severe*). Section II consists of five items that measure the degree of influence of the symptoms on school, work, usual social activity, and relationships with others. A respondent is regarded as having PMDD if they meet all of the three criteria: (1) answering *severe* for at least one of the items regarding mood symptoms in section A, (2) answering *moderate* or *severe* at least 4 items in section A, (3) answering *severe* for at least one item in section B.

2.3.2. ADHD/ASD traits

We used The Adult ADHD Self-Report Scale (ASRS-v1.1), which is a self-reported scale comprising 18 items to assess ADHD symptoms based on the diagnostic criteria of ADHD from DSM-IV (Kessler et al., 2005). The items are divided into Parts A and B. Part A contains six questions most acutely predictive of a diagnosis of ADHD, assessing the respondent's frequency of occurrence of ADHD symptom on a 5-point Likert scale: *never*, *rarely*, *sometimes*, *often*, and *very often*. We used

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Part A and categorized the participants to have ADHD traits when four or more of the six questions were applicable. The Japanese version of ASRS showed acceptable reliability and validity (Takeda et al., 2017).

As for the ASD traits, we used a 10-item Autism Spectrum Quotient (AQ10). It identifies 10 questions as potential indicators of ASD. The respondent determines the extent to which a symptom is personally applicable on a 4-point Likert scale: *definitely agree*, *slightly agree*, *slightly disagree*, and *definitely disagree*. All answers are dichotomized for processing, and they are transformed to 0 (not indicative of ASD) or 1 (indicative of ASD). This produces a scoring range from 0 to 10, with higher scores indicating more ASD traits. The cutoff score of 6 was used to have ASD traits according to the Japanese version of the AQ questionnaire. The Japanese version of AQ10 showed acceptable reliability and validity (Kurita et al., 2005).

2.3.3. Demographic variables

For demographic variables, we asked age (20–29 years [ref], 30–39 years, 40–44 years), educational attainment (junior high/high school [ref], undergraduate, or postgraduate), and occupation (non-manual work[ref] or manual work).

2.3.4. Lifestyle habits

We asked about lifestyle habits including regular breakfast, sleeping hours, exercise, and sitting hours while working. The habit of having breakfast was asked by a single question: "Do you usually have breakfast?" Sleeping hours per night were used as a continuous variable. For exercise, we asked a multiple-choice question: "Do you have a habit of sweaty exercise for more than 30 min?" The possible answers were 'None,' 'Once a month,' 'Once a week,' and 'More than twice a week." Sitting hours were assessed by a question "To what extent do you spend sitting during your work?" The possible answers ranged from 0 % to a maximum of 100 %.

2.3.5. Medication-related factors

Hospital visit history due to psychiatric illness was also used as a variable. Visits due to psychiatric illness were assessed by a single question "Have you ever visited a psychiatric hospital or mental clinic?" The possible answers were 'Yes (Currently),' 'Yes (In the past),' and 'No (Never).' The use of anti-inflammation drugs during menstruation was asked "How often do you use anti-inflammatory drugs during your menstrual period?" The possible answers were 'None,' 'Sometimes,'

Table 1
Demographic characteristics of participants (N = 2000).

	Total		Participants with PMDD (n = 101)		Participants without PMDD (n = 1899)	
	N (%)	Average (SD) [Min, Max]	N (%)	Average (SD) [Min, Max]	N (%)	Average (SD) [Min, Max]
Age (years)		33.7 (6.42) [20.0, 44.0]		32.0 (6.38) [20.0, 44.0]		33.8 (6.41) [20.0, 44.0]
20–29	603 (30.2)		40 (39.6)		563(29.6)	
30–39	915 (45.8)		47 (46.5)		868 (45.7)	
40–44	482 (24.1)		14 (13.9)		468 (24.6)	
Educational attainment (years)						
High school or less	776 (38.8)		41(40.6)		735 (38.7)	
Undergraduate	1137 (56.9)		58 (57.4)		1079 (56.8)	
Postgraduate	87 (4.4)		2 (2.0)		85 (4.5)	
Occupation						
Non-manual work						
Management	70 (3.5)		3 (3.0)		67 (3.5)	
Professional/technical profession	556 (27.8)		35 (34.7)		521 (27.4)	
Office worker	968 (48.4)		46 (45.5)		922 (48.6)	
Sales worker	226 (11.3)		8 (7.9)		218 (11.5)	
Others	50 (2.5)		2 (2.0)		48 (2.5)	
Manual work						
On-site service	130 (6.5)		7 (6.9)		123 (6.5)	
Sitting hours during working (%)		61.7 (34.2) [0, 100]		56.7 (34.4) [0, 100]		62.0 (34.2) [0, 100]
Sleeping hours(hours)		6.38 (1.03) [3.0, 12.0]		6.24 (1.07) [3.0, 8.0]		6.39 (1.03) [3.0, 12.0]
Habit of eating breakfast						
Yes	1548 (77.4)		74 (73.3)		1474 (77.6)	
No	452 (22.6)		27 (26.7)		425 (22.4)	
Regular Exercise for more than 30 min						
None	982 (49.1)		41 (40.6)		941 (49.6)	
Once a month	244 (12.2)		13 (12.9)		231(12.2)	
Once a week	379 (19.0)		24 (23.8)		355 (18.7)	
More than twice a week	395 (19.8)		23 (22.8)		372 (19.6)	
History of mental clinic/hospital visits						
Currently	137 (6.9)		18 (17.8)		119 (6.3)	
Past	234 (11.7)		26 (25.7)		208 (11.0)	
Never	1629 (81.5)		57 (56.4)		1572 (82.8)	
Use of anti-inflammation drug during menstruation						
None	718 (35.9)		13 (12.9)		705 (37.1)	
sometimes	593 (29.7)		21 (20.8)		572 (30.1)	
Always (once)	262 (13.1)		13 (12.9)		249 (13.1)	
Always (twice)	277 (13.9)		24 (23.8)		253(13.3)	
Always (more than 3 times)	150 (7.5)		30 (29.7)		120 (6.3)	
Prevalence of ADHD traits	172 (8.6)		35 (20.3)		137(79.7)	
Prevalence of ASD traits	255 (12.8)		17 (6.7)		238 (93.3)	
Prevalence of Both ASD and ADHD traits	30 (1.5)		7 (23.3)		23 (76.7)	
Total	2000 (100)		101 (5.1)		1899 (95.0)	

Abbreviations: SD, standard deviation; ADHD, Attention-Deficit/Hyperactivity Disorder; ASD, Autism Spectrum Disorder; PMDD, Pre-Menstrual Dysphoric Disorder.

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'Always (Once),' 'Always (twice),' and 'Always (more than 3 times).'

2.4. Statistical analysis

First, multivariate logistic regression analysis was performed to test whether the presence of ADHD/ASD traits could affect PMDD, by adjusting age, educational attainment, and occupation. If the relation was significant, we calculated the interaction effect to determine whether there were differences in the relationship between PMDD and the variables depending on the presence or absence of developmental traits. When the interaction effect was not significant, we conducted a stratified analysis to evaluate the association between these factors. There were no missing values because the participants could not complete the online survey when they had even one unanswered item. Therefore, we did not adopt any methods to consider potential bias due to data loss due to missing. Statistical analyses were conducted using R Version 4.2.1. A significance level of 0.05 was considered statistically significant.

3. Results

All of the participants answered the questions and were included in the final analysis. Table 1 shows the demographic characteristics of the study participants and the prevalence of PMDD. The mean age was 33.7 (standard deviation 6.42). More than half of the participants had an education higher than undergraduate school. Almost half of the participants worked as office workers. As for the lifestyle habits, about 80 % had a habit of having breakfast but half of the people did not have regular exercise. The majority of the participants had never visited psychiatric hospitals or mental clinics. During menstruation, the frequency of taking anti-inflammatory drugs varied almost evenly among our study participants.

As for developmental traits, 8.6 % had ADHD traits, whereas 12.8 % of the participants had ASD traits. The participants who had both ADHD and ASD traits were small (1.5 %). The prevalence of PMDD was 5.1 % in total. Participants with ADHD traits had a high prevalence of PMDD (20.3 %) compared to participants with ASD traits (6.7 %).

Table 2 shows the main results of our study. ADHD traits had a significant association with PMDD. The association was significant even after adjusting for covariates (adjusted OR = 6.49, 95 % CI, 4.10–10.2, $p < 0.001$). On the other hand, the association between ASD and PMDD was not significant (adjusted OR = 1.35, 95 % CI, 0.76–2.26). The association between both ADHD and ASD traits and PMDD was also significant (adjusted OR = 5.56, 95 %CI, 2.14–12.8, $p < 0.001$, Appendix 1).

Table 3 presents the interaction effect and a stratified analysis based on the presence of ADHD traits. The interaction effect was insignificant, and we could not find which variables were specifically effective in the association between ADHD traits and PMDD. The stratified analysis showed that among participants with ADHD traits, those who exercise "more than twice a week" exhibited a significantly elevated risk of developing PMDD (adjusted OR = 2.73, 95 % CI, 1.03–7.35, $p = 0.044$). Among those without ADHD traits, those who visit mental clinic/hospitals 'currently' had a significantly reduced risk of having PMDD (OR = 0.21, 95 %CI, 0.11–0.43, $p < 0.001$). Participants with ADHD traits and those without ADHD traits reported that using anti-inflammatory medications for menstrual symptoms frequently tended to exhibit significantly higher risk of having PMDD.

4. Discussion

This study explored the association between ADHD/ASD traits and PMDD in a cohort of working women who had not experienced pregnancy. The results revealed that ADHD traits had a significant association with PMDD, which was in line with our hypothesis. However, no significant association was found between ASD traits and PMDD. The interaction effect was insignificant in identifying specific variables influencing the association between ADHD traits and PMDD. The stratified analysis revealed that frequent exercise increased the risk of PMDD among individuals with ADHD traits, while current mental health treatment reduced the risk among those without ADHD traits. Frequent use of anti-inflammatory medications was associated with a higher risk of PMDD in both groups.

The significant association between ADHD traits and PMDD observed in this study aligns with results from previous research (Dorani et al., 2021; Lin et al., 2024). The present study limited participants who were female workers with no history of contraceptive use or pregnancy. The results suggested a potential biological etiology and mechanism of PMDD: normal ovarian function acts as a cyclical trigger for biochemical events associated with PMDD and increased sensitivity to fluctuations in gonadal hormones may contribute to menstrual-related symptoms, with dysregulation of the serotonergic system identified as a key factor. This likely explains the efficacy of selective serotonin reuptake inhibitors in treating PMDD (Steiner and Perlstein, 2000; Halbreich, 2003; Meyer, 2023). ADHD, on the other hand, is associated with dysregulation in intracellular signaling systems and reduced dopamine levels in several brain regions, potentially leading to mood disturbances (Wu et al., 2012). This raises the possibility that the interaction between sex

Table 2
Association between ADHD/ASD traits and PMDD in adult women population of full-time employees in Japan (N = 2000).

Dependent variable (PMDD)								
	Independent variable:ADHD				Independent variable:ASD			
	Crude		Adjusted		Crude		Adjusted	
	OR (95 %CI)	*p*	OR (95 %CI)	*p*	OR (95 %CI)	*p*	OR (95 %CI)	*p*
ADHD/ASD traits [ref: Negative]	6.82 (4.33–10.6)	<0.001	6.49 (4.10–10.2)	<0.001	1.41 (0.80–2.36)	0.209	1.35 (0.76–2.26)	0.277
Covariates								
Age 20–29	–	–	1(ref)	–	–	–	1(ref)	–
30–39	–	–	0.90 (0.58–1.42)	0.653	–	–	0.77 (0.50–1.20)	0.245
40–44	–	–	0.55 (0.28–1.01)	0.064	–	–	0.43 (0.22–0.79)	0.009
Educational attainment High school or less	–	–	1 (ref)	–	–	–	1 (ref)	–
Undergraduate	–	–	0.91(0.60–1.41)	0.674	–	–	0.93(0.62–1.42)	0.729
Postgraduate	–	–	0.40 (0.06–1.37)	0.218	–	–	0.44 (0.07–1.47)	0.264
Occupation [ref: Manual work]	–	–	0.98 (0.46–2.44)	0.966	–	–	0.96 (0.46–2.33)	0.916

Abbreviations: ADHD, Attention-Deficit/Hyperactivity Disorder; ASD, Autism Spectrum Disorder; OR, odds ratio; CI, confidence intervals; PMDD, Pre-Menstrual Dysphoric Disorder.
Note: Multiple logistic regression was performed showing odds ratios and 95 % confidence intervals for PMDD. Adjusted model adjusted for age, educational attainment, and occupation.

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Table 3
Association between lifestyle habits/medical related factors and PMDD among ADHD traits positive and negative women of full-time employees in Japan (N = 2000).

	ADHD traits positive (n = 172)			ADHD traits negative (n = 1828)			Group difference [b]
Variables [a]	N (%)	OR (95 %CI)	*p*	n	OR (95 %CI)	*p*	*p*
Sitting hours during worktime	172 (100)	1.01 (0.99–1.02)	0.365	1828 (100)	0.99 (0.99–1.00)	0.040*	0.065
Sleeping hours	172 (100)	0.93 (0.65–1.33)	0.691	1828 (100)	0.86 (0.67–1.09)	0.203	0.704
Habit of eating breakfast [ref: No]	120 (69.8)	0.93 (0.43–2.14)	0.863	1428 (78.1)	0.87 (0.50–1.59)	0.637	0.893
Regular exercise None	72 (41.9)	1 (ref)	–	910 (49.8)	1 (ref)	–	–
Once a month	30 (17.4)	1.24 (0.36–3.87)	0.719	214 (11.7)	1.10 (0.47–2.32)	0.811	0.869
Once a week	34 (19.8)	2.23 (0.80–6.20)	0.120	345 (18.9)	1.29 (0.67–2.38)	0.429	0.367
More than twice a week	36 (20.9)	2.73 (1.03–7.35)	0.044*	359 (19.6)	0.98 (0.48–1.88)	0.955	0.091
History of visiting mental clinic/hospitals None	122 (70.9)	1.0 (ref)	–	1507 (82.4)	1 (ref)	–	–
Past	28 (16.3)	1.07 (0.31–3.84)	0.919	206 (11.3)	0.82 (0.38–1.81)	0.617	0.211
Currently	22 (12.8)	0.55 (0.20–1.69)	0.270	115 (6.3)	0.21 (0.11–0.43)	<0.001*	0.129
Use of anti-inflammatory drugs during menstruation None	43 (25.0)	1(ref)	–	675 (36.9)	1 (ref)	–	–
Sometimes	48 (27.9)	3.51 (0.99–16.5)	0.071	545 (29.8)	1.37 (0.57–3.31)	0.475	0.254
Always(once)	32 (18.6)	1.90 (0.39–10.3)	0.422	230 (12.6)	2.71 (1.06–6.80)	0.033*	0.705
Always (twice)	30 (17.4)	10.2 (2.86–48.8)	<0.001*	247 (13.5)	3.10 (1.29–7.53)	0.011*	0.152
Always (more than 3 times)	19 (11.0)	4.76 (1.04–25.8)	0.049*	131 (7.2)	15.7 (7.53–35.1)	<0.001*	0.177

Abbreviations: ADHD, Attention-Deficit/Hyperactivity Disorder; OR, odds ratio; CI, confidence intervals; PMDD, Pre-Menstrual Dysphoric Disorder.
[a] Variables were input individually. Univariate logistic regression was performed showing odds ratios and 95 % confidence intervals for PMDD.
[b] The interaction effect was calculated to determine whether there were differences in the relationship between PMDD and the variables depending on the presence or absence of ADHD.

hormones and molecules involved in synaptic plasticity and neurotransmitter systems could contribute to clinical characteristics in women with ADHD. Such mechanisms might influence gonadal function and alter hormonal fluctuations across the menstrual cycle (Haimov-Kochman and Berger, 2014). From an epidemiological perspective, this study highlights the significant association between ADHD traits and PMDD, warranting further investigation into shared and distinct mechanisms.

In contrast, no significant association was found between ASD traits and PMDD. This result was in line with the previous study that reported no significant increase in the prevalence of PMDD among individuals with ASD (Groenman et al., 2022). Autism spectrum conditions are more commonly observed in males, and historically fewer studies have focused on ASD in women. As a result, little is known about how women with ASD experience and cope with hormonal changes, including menopause. However, one qualitative study found that menopausal experiences varied greatly among women with ASD, with some reporting significant declines in daily functioning, coping skills, mental health, and social engagement (Moseley et al., 2021). Further epidemiological studies with biological implications for hormone sensitivity in women with ASD are necessary.

The association between both ADHD/ASD traits and PMDD was also significant. ADHD and ASD share environmental and biological risk factors; however, research on this topic remains limited, particularly among adults with comorbid ADHD and ASD (Antshel et al., 2016). The odds ratio remains consistent with the results accounting for ADHD and PMDD, suggesting that ADHD itself might have influenced the outcome. In our study, the number of participants exhibiting both ADHD and ASD traits was small. Further research is necessary to determine whether the mechanisms underlying the positive association with PMDD partially overlap between ASD and ADHD traits.

Although the interaction effect was not significant, several potential factors influencing the association between ADHD traits and PMDD can be inferred from our results. First, this study showed that frequent exercise was associated with an increased risk of PMDD among individuals with ADHD traits. Due to ADHD traits such as hyperactivity and hyperfocus the individual may have a reduced perception of fatigue in response to physical exertion, which could potentially exacerbate PMDD symptoms. Although regular exercise is generally considered protective for overall health (Liguori et al., 2023), exercise including competitive sports has been found to promote PMDD (Czajkowska et al., 2019), and the effectiveness of exercise as a coping strategy for PMDD may be inherently limited. Therefore, the finding that high-frequency exercise may exacerbate PMDD symptoms warrants careful consideration. Factors such as the type and duration of exercise should also be examined in future research. Second, current mental health treatment was associated with low odds of PMDD among individuals without ADHD traits. Since PMDD is influenced by psychological factors such as depressive symptoms, it is theoretically expected that the odds ratios would be higher. However, psychotherapy or pharmacotherapy targeting psychiatric symptoms may have incidentally alleviated menstrual symptoms, resulting in lower odds ratios. Future studies should take psychiatric disorders into consideration. Finally, our results indicated that those who reported using anti-inflammatory medications for menstrual symptoms had a significant positive association with PMDD, regardless of ADHD traits (Table 3). This suggests that the use of analgesics may reflect noticeable menstrual-related symptoms, prompting individuals to self-medicate. However, as the primary treatments for PMDD typically involve antidepressants or low-dose oral contraceptives, the reliance on painkillers may indicate that female workers with PMDD are not receiving appropriate medical care for their condition. Given that approximately 30 % of individuals with ADHD traits have visited a psychiatric hospital or clinic in the past or are currently receiving care (Appendix 2), they have opportunities to receive treatment for PMDD. If PMDD were properly recognized in these cases, appropriate treatment could alleviate the burden of menstrual-related symptoms. Thus, collaboration with psychiatric care providers is essential, particularly in addressing menstrual-related symptoms during consultations for individuals with ADHD. Moreover, women with PMDD are more likely to experience comorbid conditions such as dysmenorrhea (Kitamura et al., 2012) and menstrual migraines (Barone et al., 2023), as these conditions share pathophysiology through fluctuations in female hormones and influence psychological stress. Our result may suggest the existence of female workers who are unaware of having PMDD or who are managing the symptoms with painkillers. Leading these workers to appropriate standard treatments for PMDD may also be effective in alleviating these pain-related conditions. Proper self-care guidance providing comprehensive treatment may be crucial in this context. To note, more than 70

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% of individuals with ADHD traits lack access to psychiatric care. To reduce the burden of menstrual-related symptoms of female workers, raising awareness of ADHD traits to screen for PMDD among occupational healthcare workers may also be a key component in identifying potential interventions in workplaces or clinical settings.

5. Limitations

This study has several limitations. First, cross-sectional design limits our ability to assess causal relationships, although PMDD is unlikely to affect ADHD/ASD traits. PMDD is diagnosed based on prospective daily symptom ratings across at least two menstrual cycles, while developmental disorders are characterized by symptom changes across different stages of life. The cross-sectional design also restricts our ability to capture temporal fluctuations in these conditions, which are marked by distinct time courses and symptom phases. Longitudinal research is required to establish causal relationships and confirm these associations. Notably, the observed correlations between ADHD traits and PMDD may be partially explained by shared attentional symptoms. It remains unclear whether attention-related difficulties are confined to the late luteal phase or persist throughout the entire menstrual cycle. In our study, we were unable to assess fluctuations in attentional symptoms in relation to menstrual cycle phases. Addressing this limitation should be a priority in future research. Second, while we controlled for potential confounding variables (age, education, work), the possibility of unmeasured confounding variables cannot completely be ruled out. For example, medical history of mood disorder and other psychiatric disorders including anxiety symptoms or sexual traumatic experience in childhood may affect the association. Third, all data were obtained through self-reported questionnaires. We screened ADHD/ASD traits by self-reporting questionnaire, which differs from the clinical diagnosis of developmental disorders. In particular, the proportion of participants with ASD traits was higher than the global prevalence of adult ASD reported by previous study (Hirota and King, 2023). Caution is warranted as the use of a self-administered questionnaire may have led to overestimation. Moreover, the participants in this study were limited to full-time employees; however, individuals with ADHD/ASD struggle to maintain stable employment (Roux et al., 2013; Sayal et al., 2018). The demands of full-time work might have contributed to a higher burden of work and exacerbation of PMDD, potentially leading to an overestimation of the odds ratio. As for PMDD, the prevalence of having PMDD in our result was within the current estimation of the prevalence for PMDD. However, the screening tool we used for PMDD is different from the formal diagnostic procedure that usually records prospective symptoms over two menstrual cycles. Furthermore, as the data was collected through a self-administered questionnaire, it should be noted that this does not equate to a clinical diagnosis. Moreover, the screening tool is based on the diagnostic criteria of the DSM-IV-TR, rather than the updated criteria outlined in the DSM-5-TR. In future studies, we should consider controlling for menstrual cycle timing based on the diagnostic criteria outlined in the DSM-5-TR. Fourth, the recruitment of participants through an online platform may limit the generalizability of the results due to potential selection bias. Despite these limitations, this is the first study to focus exclusively on full-time working women who have not experienced pregnancy and are not using contraceptives. This is a key strength, as it allowed us to recruit a large and homogenous sample of female workers under similar working, biological, and medical conditions. Such uniformity allowed the researchers to examine the association between ADHD/ASD traits and PMDD while minimizing potential confounding factors, such as hormonal changes due to pregnancy or the therapeutic effects of contraceptive use, both of which are known to influence PMDD symptoms. The results provide valuable insights for healthcare professionals to enhance awareness of ADHD/ASD traits and PMDD, particularly among individuals who may not frequently access healthcare services.

6. Conclusion

Among full-time female employees, a significant association was found between ADHD traits and PMDD, while no significant association was observed between ASD traits and PMDD. Women with ADHD traits may require special attention for the treatment of PMDD and additional support in managing menstrual-related symptoms.

CRediT authorship contribution statement

Rikako Tsuji: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kazuhiro Watanabe:** Writing – review & editing, Supervision, Methodology. **Miho Egawa:** Writing – review & editing, Methodology, Investigation. **Yuka Ito:** Writing – review & editing, Investigation. **Yoshiaki Kanamori:** Writing – review & editing, Investigation. **Mako Iida:** Writing – review & editing, Investigation. **Daisuke Nishi:** Writing – review & editing, Supervision, Methodology, Investigation. **Natsu Sasaki:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Informed consent statement

Online informed consent was obtained from all participants, who were given full disclosure and explanation of the purpose and procedures of this study. We explained that their participation was voluntary, and they could withdraw consent for any reason simply by not completing the questionnaire.

Institutional review and board statement

The study was conducted according to the guidelines of the Declaration of Helsinki and was approved by the Research Ethics Committee of the Graduate School of Medicine/Faculty of Medicine, The University of Tokyo, No. 2023058NI-(1).

Data availability statement

The data presented in this study are available from the corresponding author, NS, upon reasonable request.

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Declaration of competing interest

Natsu Sasaki received personal fees from Medilio Co., Ltd., outside the submitted work. DN reports personal fees from Startia, Inc., MD.net, and an honorarium from Takeda Pharmaceutical Co., Otsuka Medical Devices Co., Ltd. outside the submittedwork. The other authors state that they have no conflicts of interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

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